Synera

The AI Agent Platform for Engineering

CHALLENGE BRIEF · INDUSTRIAL AGENTS HACKATHON 2026

Talk to Your Building: Agent-First Indoor Navigation

The story

You have walked into an office complex or a factory hall and had no idea where to go. Outdoors we solved this long ago: type where you want to be, and a map shows the way. Indoors, floor plans are PDFs, machines live in spreadsheets, and the one person who knows where the 3D printer is has gone home.

Your mission: give any building its own map you can talk to. Feed your system a floor plan and it learns the building. Then anyone can just talk to it.

Questions your agent should answer

1. **FIND** "Where is the 3D printer?"
2. **ROUTE** "How do I get from the lobby to the CNC mill on floor 2 without stairs?"
3. **EDIT** "Add a door between rooms 2.03 and 2.04."
4. **REACT** "The east lift is closed for maintenance. What changes?"

About Synera

Synera is agentic AI for engineering, open and connected across every tool in the product development stack. Our AI agents work as digital engineering coworkers that change geometry, run simulations and iterate, at companies such as NASA, Airbus and BMW. This challenge is a compact version of what we work on every day: agents that understand space and geometry and explain themselves to people in plain language.

What you build

An app you talk to. It shows 2D or 3D views of buildings and floors, animated routes, and the things inside: rooms, desks, machines, equipment. It works for an existing building and can set up a new one quickly from floor plans.

The agent is the interaction layer: you say what you want and the system does it. The core underneath is mostly deterministic: a graph of rooms, doors, corridors, stairs and lifts, and a search that finds the best route.

What you get

All data is in GitHub repository: Synera-GmbH/industrial-agents-hackathon-2026. No company data and no NDA; everything is public or generated.

- **Floor plans** of several buildings, e.g. a factory hall with offices and a four-storey campus building, as structured plan files plus PNG images. [FORMAT, e.g. JSON; the schema is in the kit.]

- **Equipment lists** per building, so there is something to find.
- **A test set** of example questions with expected answers, to check your agent while you work.
- **Credits** for Claude, Lovable and ElevenLabs via the organizers. UX and voice can play a real role: a voice in, voice out agent counts towards Interaction.

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Levels

Every level earns points and builds on the one before, so a team that stops early still has something that works.

1. **Locate:** on one floor, talk to the agent, find things and see them on a view.
2. **Route:** find the way from A to B on one floor, with options such as step-free routes, closed areas and multi-stop tours.
3. **Scale:** multiple floors, then multiple buildings, with stairs, lifts and walkways between them.
4. **Build:** create or update a building from floor plans, by conversation. A new building should take minutes, not days.
5. **Extend:** pick an add-in below, or surprise us.

Optional add-ins

For teams that finish early, and worth creativity points. Make things move: tools or machines that change place via barcode, RFID or “this is now here”, so the map stays current. Let several colleagues update the building while others ask questions. Route forklifts and materials through a hall with blocked aisles and delivery points. Capture a new building by walking through it with a phone instead of starting from a floor plan. Plan for accessibility with wider doors, step-free and quiet routes. Present it well with isometric stacked floors, animated routes and voice in and out. Or go for AR: the route as an overlay on your phone, where the hard part is knowing where you are (QR codes at doors, WiFi, other tricks). Pick one and do it well rather than three badly.

Directions, not instructions

How you get there is up to you. A few directions that save hours:

- **Think in a graph.** Rooms and junctions are nodes; doors, corridors, stairs and lifts are edges; Dijkstra or A* finds the route. Agree on the data model in your first hour.
- **One floor first.** Make one question work end to end before 3D or a pretty interface. A flat 2D plan with the route drawn on it is a fine start; three.js gets you extruded floors later.
- **Spend your time on decisions.** AI writes most of the code, so think about the user: what should the agent ask when a request is vague? How does it show what it understood before it changes a plan? “Turn left at the printers” or room numbers?

Sunday benchmark: how we run your agent

Shortly before code freeze (TODO: exact time) we release one new building nobody has seen, plus a question file. Your agent has to answer them, including the best routes.

1. **Input:** the new building in the same plan format as the training data, and questions.json with one entry per question (id, text, optional constraints such as step-free).
2. **Your run:** a single command in your repo, documented in BENCHMARK.md, e.g. run_benchmark <building_dir> <questions.json> <answers.json>. No manual steps between input and output.
3. **Output:** answers.json, one entry per question id with the located item or room id, the route as an ordered list of node ids, and the route length in metres.
4. **Submission:** you run it yourselves and submit answers.json with your project. We may re-run your command on the same input to verify.

We score correct answers, valid routes and route length against the best route. A schema and an example answer file are in the data folder (TODO).

Submission and judging

There are no pitch rooms. All teams submit online by code freeze through the organizers' platform:

- a demo video of your app in action
- a short technical explanation (architecture and tech stack)
- your repository, including BENCHMARK.md and your answers.json

We review all Track 01 submissions and pick **four finalists**. Finalists present at the joint final pitch session at Tacto: **3 minutes plus 1 minute Q&A** per team.

Three things count, in this order: does it work, is it a pleasure to use, and what did you add that we did not ask for.

Part	Points	What we look at
Does it work	55	Sunday benchmark plus automatic checks on buildings and questions you have not seen, covering levels 1 to 4: correct answers, valid routes, route length against the best route, correct buildings built from plans
Interaction	25	A short scripted conversation with tricky users, rated on a rubric. Does it ask only what it needs, show what it understood, and give directions a person can follow? Voice counts here.
Creativity	10	Level 5 and add-ins: anything that makes the product better in a way we did not think of
Submission demo	10	Your demo video and technical explanation: clear, convincing, and showing the real product